MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, M151
PARTS FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the quality assurance provisions, and special requirements not covered by the drawings, for one type of warhead designated as Warhead, 2.75 Inch Rocket, M151.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

FEDERAL

FED-STD-151	Metals, Test Methods

FSC 1340

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105	Sampling procedures and Tables for Inspection by Attributes
MIL-STD-1167	Ammunition Data Cards

2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

A-395 Standard Specification for Ferritic Ductile Iron Pressure Retaining Castings for use at elevated temperatures.

(Application for copies of ASTM publications should be addressed to The American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

DRAWINGS

US ARMY ARMMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGE DRAWINGS

8837835	Pallet for Projectiles and Projectile Metal Parts
8889406	Warhead, 2.75 Inch Rocket M151 Metal Parts Assembly

INSPECTION EQUIPMENT DRAWINGS

ET 8889406	Equipment Tabulation

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Warhead assemblies and parts. The warhead assemblies and parts shall comply with all requirements specified on drawing (Dwg) 8889406 and associated drawings, and with all requirements specified in applicable standards and specifications.

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3.3 Metal defects. The components shall be free of surface defects, cracks, pits, cold shuts, hot tears, blow holes, scaling, and other similar defects.

3.4 Hardness test.

3.4.1 Nose, malleable iron. Each nose casting from a sample shall be subjected to a hardness test with a Brinell Hardness Tester. The sample, 10 percent minimum, representative of extreme furnace environment, shall be selected from each heat treat batch when a batch type furnace is used for final heat treatment. When a continuous type furnace is used for final heat treatment, from which a 10 percent minimum sample will also be selected, a heat treat batch will be considered a quantity of castings not to exceed 4 hours production. The castings which constitute the sample shall be collected at regular continuous intervals throughout the 4 hour cycle. The heat treat batch thus constituted shall form the basis for selecting sample castings for the tension test hereinafter prescribed. The test shall be performed as specified in 4.5.2 using equipment in accordance with 4.4.4.

3.4.2 Nose, ductile iron. Each nose casting from a sample shall be subjected to a hardness test with a Brinell Hardness Tester . The-sample shall be 1.5 percent minimum, representative of extreme furnace conditions. When a continuous type furnace is used for final heat treatment, a sample shall be 1.5 percent of the castings completing heat treatment. If boxes or baskets are used which contain approximately 70 castings, one casting from each box shall be tested. A heat treat batch will be considered a quantity of castings not to exceed 8 hours production. The test shall be performed as specified in 4.5.2 using equipment in accordance with 4.4.4. The lot may be reheat treated to attain specifications and drawing requirements.

3.4.3 Base, malleable iron. Each base casting from a sample shall be subjected to a hardness test with a Brinell Hardness Tester . The sample, 10 percent minimum, representative of extreme furnace environment, shall be selected from each heat treat batch when a batch type furnace is used for the final heat treatment. When a continuous type furnace is used for final heat treatment, from which a 10 percent minimum sample will also be selected, a heat treat batch will be considered a quantity of castings not to exceed 4 hours production. The castings which constitute the sample shall be collected at regular continuous intervals throughout the 4 hour cycle. The heat treat batch thus constituted shall form the basis for selecting sample castings for the tension test hereinafter

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prescribed . The hardness test shall be performed on the approximate center of the bottom of the castings. The test shall be performed as specified in 4.5.2 using equipment in accordance with 4.4.4.

3.5 Tension test.

3.5.1 Nose, malleable iron. Two nose castings from each heat treatment batch, one of the hardest and one of the softest as determined by 3.4.1, shall be subjected to this test. The test shall be performed as specified in 4.5.3 using equipment in accordance with 4.4.4.

3.5.2 Nose, ductile iron. To qualify the heat treatment, one 0.500 ± .010 inch diameter test specimen machined from a 1 inch knock-off type keel block representative of each heat treat lot shall be subjected to the test specified in 4.5.3 using equipment in accordance with 4.4.4. In the case of the continuous heat treat oven, one specimen shall be tested for each eight (8) hours of oven operating time.

3.5.3 Base. When malleable iron is used, two base castings from each heat treat batch, one of the hardest and one of the softest as determined by 3.4.3, shall be subjected to the tension test in accordance with 4.5.4, to determine compliance with the mechanical properties requirements of the applicable drawing. When steel bases are used, two bases from each heat treatment or stress relieved batch of 5000 or fraction thereof shall be subjected to the tension test. The test shall be performed as specified in 4.5.4 using equipment in accordance with 4.4.4.

3.6 Microexamination. When ductile iron is used, a test lug shall be cast representing the last metal poured from each ladle treated with the modularizing agent. The test casting and microexamination shall be in accordance with the microstructure test described in ASTM-A-395.

3.7 Hydrostatic test. Each metal parts assembly shall be subjected to the required constant internal hydrostatic pressure to determine compliance with drawing requirements. Metal parts assemblies which comply shall be stamped with the letter “H” as indicated on the assembly drawing. The test shall be performed as specified in 4.5.7 using equipment in accordance with 4.4.4.

3.8 Air pressure. Each body assembly that has passed the hydrostatic test, as indicated by the letter “H” stamped on the body, shall after completion of the drying cycle be subjected to a

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submerged air pressure test. Body assemblies which comply with the drawing requirements shall be stamped with the letter "A" as indicated on the assembly drawing. The test shall be performed as specified in 4.5.8 using equipment in accordance with 4.4.4.

3.9 Salt spray test. The test shall be performed as specified in 4.5.9 using equipment in accordance with 4.4.4.

3.10 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by-the contractor shall be as specified in the contract.

3.11 Workmanship. All parts and assemblies shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface blisters, excess flash, dirt, greese, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for minor scratches not exposing base material. The assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Nose	8889405	5 (Unmachined) 5 (Machined)
Base	8889404	5 (Unmachined) 5 (Machined)
Warhead, 2.75 Inch Rocket, M151, Metal Parts Assembly	8889406	5 (Unpainted) 5 (Painted)

Tensile specimens:

a. Base extrusion: Two (2) "R" type specimens; one from each of two unmachined extrusions. Carcasses from which the specimens are cut shall accompany the specimens.

b. Nose casting: Four (4) "R" type tensile specimens; two from each of two unmachined castings. The two specimens shall be cut from the straight portion of the casting forward of the bourrelet and 180 degrees apart.

c. Ductile iron keel block: Two (2) 0.500 + .010 inch diameter machined test specimens; one from each of two 1 Inch knock-off type keel blocks.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein. Parenthetic numbers in Table I refer to the equipment lists cited in ET 8889406.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-W-50504A (AR)

PARAGRAPH	TITLE: Warhead, 2.75 Inch Rocket, M151, Metal Parts Assembly		SHEET 1 OF 1		DRAWING NUMBER: Listed Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Nose (Dwg. 8889405)				
	Examination for defects	10(a)		3.2	4.4.2.1
	Hardness test	(b)		(b)	4.5.2 (01002 or 01003)
	Tension test	4		3.5.1	4.5.3 (01004 or 01005)
	Ductile iron keel block tension test	2		3.5.2	4.5.3
	Microexamination	(c)		3.6	4.5.6
	Base (Dwg. 8889404)				
	Examination for defects	10(a)		3.2	4.4.2.2
	Hardness test	(d)		3.4.3	4.5.2 (02007)
	Tension test	2		3.5.3	4.5.4 (02008)
	Warhead, 2.75 Inch Rocket, M151 Metal Parts Assembly (Dwg. 8889406)				
	Examination for defects	10(e)		3.2	4.4.2.3/4.4.2.4
	Hydrostatic test (unpainted)	5		3.7	4.5.7 (03028)
	Air pressure test (unpainted)	5		3.8	4.5.8 (03029)
	Salt spray test (painted)	5		3.9	4.5.9 (04013)

NOTES:
(a) 5 unmachined, 5 machined.
(b) See 3.4.1 or 3.4.2, whichever is applicable.
(c) See 3.6 (ductile iron only).
(d) See 3.4.3.
(e) 5 unpainted, 5 painted.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8889405
4.4.2.1	Nose				NEXT HIGHER ASSEMBLY 8889406
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Metal defective		0.40%	3.3	Visual
102	Hardness test	(a)		(a)	4.5.2 (01002 or 01003)
103	Tension test	(b)		(b)	4.5.3/4.5.5 (01004 or 01005)
104	Microexamination	(c)		3.6	4.5.6
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:
(a) See 3.4.1 or 3.4.2, whichever is applicable.
(b) See 3.5.1 or 3.5.2, whichever is applicable.
(c) See 3.6 (ductile iron only).

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8889404
4.4.2.2	Base				NEXT HIGHER ASSEMBLY 8889406
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Inside diameter at front		0.40%	3.2	Gage (02001)
102	Inside diameter		0.40%	3.2	Gage (02002)
103	Metal defective		0.40%	3.3	Visual
104	Hardness test	(a)		3.4.3	4.5.2 (02007)
105	Tension test	(b)		3.5.3	4.5.4/4.5.5 (02008)
Minor					
201	Improper radii or chamfer		0.65%	3.2	Visual/Gage
202	Improper surface finish		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:
(a) See 3.4.3.
(b) See 3.5.3.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50504A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Warhead, 2.75 Inch Rocket, M151, Metal Parts Assembly, before painting				8889406
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Hydrostatic test		100%	3.7	4.5.7 (03028)
2	Air pressure test		100%	3.8	4.5.8 (03029)
Major					
101	Concentricity of pitch diameter of fuze thread		0.40%	3.2	Gage (03003)
102	Concentricity of counterbore at front of fuze thread		0.40%	3.2	Gage (03004)
103	Concentricity of diameter at rear of fuze thread		0.40%	3.2	Gage (03005)
104	Concentricity of ogive		0.40%	3.2	Gage (03006)
105	Concentricity of base at rear of threads with pitch diameter		0.40%	3.2	Gage (03007)
106	Concentricity of base at rear of threads with outside diameter		0.40%	3.2	Gage (03008)
107	Concentricity of bourrelet		0.40%	3.2	Gage (03009)
108	Perpendicularity of motor body seat		0.40%	3.2	Gage (03010)
109	Length of fuze thread		0.40%	3.2	Gage (03011)
110	Diameter of counterbore at front fuze thread		0.40%	3.2	Gage (03012)
111	Diameter of rear of fuze thread		0.40%	3.2	Gage (03013)
112	Diameter of nose between bourrelet and ogive		0.40%	3.2	Gage (03014)
113	Diameter of base at rear of threads		0.40%	3.2	Gage (03015)

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50504A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.3	Warhead, 2.75 Inch Rocket, M151, Metal Parts Assembly, before painting		8889406
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
114	Thickness of wall between bourrelet and ogive		0.40%	3.2	Gage (03016)
115	Thickness through base		0.40%	3.2	Gage (03019)
116	Length from front face to motor housing seat		0.40%	3.2	Gage (03021)
117	Length from motor housing seat to rear of base		0.40%	3.2	Gage (03022)
118	Variation of wall thickness excessive		0.40%	3.2	Gage (03017)
119	Metal defective		0.40%	3.3	Visual
120	Profile ogive		0.40%	3.2	Gage (03018)
Minor					
201	Depth of counterbore at front		0.65%	3.2	Gage (03023)
202	Improper radii or chamfer		0.65%	3.2	Visual
203	Location of rear thread		0.65%	3.2	Gage (03027)
204	Improper surface finish		0.65%	3.2	Tactual/Visual
205	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50504A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8889406
4.4.2.4	Warhead, 2.75 Inch Rocket, M151, Metal Parts Assembly, after painting				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Pitch diameter of fuze thread		0.40%	3.2	Gage (04001)
102	Minor diameter of fuze thread, max.		0.40%	3.2	Gage (04002)
103	Pitch diameter of rear thread		0.40%	3.2	Gage (04003)
104	Major diameter of rear thread, min.		0.40%	3.2	Gage (04004)
105	Thread damaged or not full		0.40%	3.2	Visual
106	Foreign matter in cavity		0.40%	3.2	Visual
107	Diameter of bourrelet, max.		0.40%	3.2	Gage (04007)
108	Salt spray test	(a)		3.9	4.5.9
Minor					
201	Protective coating inadequate or damaged		0.65%	3.2	Visual
202	Paint in cavity inadequate		0.65%	3.2	Visual
203	Paint not dry		0.65%	3.2	Tactual
204	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

(a) See 4.5.9.

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4.4.3 Testing. Inspection lots shall meet the requirements of the tests called out in the Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1; parenthetic numbers refer to the equipment lists cited on ET 8889406. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein. The contractor shall submit all inspection equipment designs to the government for evaluation and approval prior to fabrication of such equipment.

4.5 Methods of inspection.

4.5.1 Materials, components, and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.2 Hardness test. After final heat treatment, the hardness test shall be performed on the nose (forward of the bourrelet) and on the center of the base. Procedures for these tests are as prescribed by FED-STD-151, Method 242.1. If any ductile iron specimen is outside the hardness limits, the lot shall be rejected.

4.5.3 Tension test on nose. Tension test shall be performed on specimen as prescribed in FED-STD-151, Method 211.1. If test specimen is visibly defective, another specimen will be tested from the same heat treat lot. Failure of one or more tension test specimens to comply with the physical requirements of the applicable drawing shall be cause for rejection of the batch represented by the specimen.

4.5.4 Tension test on base. From the samples provided, one tensile test specimen conforming to a Standard "R" type specimen as prescribed by FED-STD-151 shall be taken from the bottom of each base. The criteria for determining yield strength and elongation shall be as prescribed by FED-STD-151, Method 211.1. Failure of any specimen to comply with the applicable drawing shall be cause for rejection of the batch represented by the specimen.

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4. 5.5 Retest.

4.5.5.1 Malleable iron. When one or more tension test specimens fail to comply with the yield strength or elongation requirements, the rejected heat treat batch may, at the discretion of-the contractor, be reheat-treated, in which case the samples shall be selected in accordance with the hardness test requirements. In the event a reheat is not deemed necessary, each nose casting of the heat treat batch shall be tested in accordance with 4.5.2 to establish the hardest and softest nose castings to be tested in accordance with 4.5.3. When maximum and minimum hardness values have been determined by compliance of the tensile test specimens with the mechanical requirements, these values shall be used as a basis for accepting or rejecting each nose casting within the heat treat batch. All castings within the heat treat batch with hardness values outside the established range shall be rejected.

4.5.5.2 Ductile iron. When the tension test specimen fails to comply with the yield strength or elongation requirements, the contractor may elect to reheat-treat and the rejected quantity may be retested by selection of samples in accordance with the hardness test requirements. If any specimen is outside hardness limits, the lot shall be rejected. The hardest and softest sample shall be tested in accordance with 4.5.3. If one or more tensile specimens fail to comply with the mechanical properties requirements, the lot shall be rejected. In the event a reheat is not deemed necessary, each nose casting of the day’s heat treat production shall be tested in accordance with 4.5.2 to establish the hardest and softest nose castings to be tested in accordance with 4.5.3. When maximum and minimum hardness values have been determined by compliance of the tensile test specimens with the mechanical requirements, these values shall be used as a basis for accepting ‘or rejecting each nose casting within the quantity. All castings within the quantity whose hardness values fall outside the established range shall be rejected.

4.5.5.3 Malleable iron base castings. When one or more tension test specimens fail to comply with the yield strength or elongation requirements, the rejected heat treat batch may at the discretion of the contractor be reheat-treated, in which case the samples shall be selected in accordance with the hardness test requirements. In the event a reheat is not deemed necessary, each base casting of the heat treat batch shall be hardness tested in accordance with 4.5.2 to establish the hardest and softest base castings to be tested in accordance with 4.5.4. When maximum and minimum hardness values have been determined by compliance of the tensile test specimens with the

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mechanical requirements, these values shall be used as a basis for accepting or rejecting each base casting within the heat treat batch. All castings within the heat treat batch whose hardness values fall outside the established range shall be rejected.

4.5.6 Microexamination. Perform a microexamination of the test lug in accordance with ASTM-A-395. Failure of the test lug to conform to the microstructure requirements of ASTM-A-395 shall be cause for rejection of the pouring batch.

4.5.7 Hydrostatic test. The metal parts assembly shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached, after which the pressure will be held constant for a minimum of 5 seconds. The equipment shall be calibrated prior to the start of each day's operation, at intervals of not over 2 hours of continuous operation, and prior to restarting the operation after a layoff of 1 hour or more. Any assembly that fails to comply with the drawing requirement shall be classed defective and removed from the lot.

4.5.8 Air pressure test. Body assemblies that have passed the hydrostatic test shall be mounted in an approved fixture in such a manner that the body assembly will be completely submerged in water. The air pressure shall be applied and held for a minimum of 15 seconds. Any body assembly failing to comply with the drawing requirements as evidenced by air bubbles rising to the surface of the water shall be classed defective and removed from the lot. Air bubbles caused by entrapped air on the outside of the shell shall not be cause for rejection of the body assembly.

4.5.9 Salt spray test. An accelerated corrosion test shall be performed utilizing the equipment and procedures prescribed by FED-STD-151, Method 811. A 20 percent concentration of sodium chloride solution shall be used and length of test shall be for 144 hours. Actual production warhead assemblies only shall be subjected to this test. Test panels shall not be substituted for the warhead assemblies. The test specimen shall be air dried for a minimum of 96 hours before testing. The painted specimen shall be scored through using a sharp knife. The specimens shall then be exposed to the 20 percent salt spray as specified in FED-STD-151, Method 811. The sample size for this test shall be as follows: During production, one warhead from each days production shall be subjected to the salt spray test until five consecutive days production have successfully

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passed the test. Sampling shall then be reduced to one sample twice per week. If upon reduced sampling one failure occurs, corrective action shall be taken and sampling shall revert to one specimen from each days production until five consecutive days' production have again successfully passed the test. If two consecutive failures occur after the contractor has been notified to take corrective action in either the original sampling plan or the reduced sampling plan, inspection for acceptance will be stopped until necessary corrective action has been taken. Any specimen (warhead) which does not comply with any of the following test criteria shall be considered a failure. The specimen, when subjected to the salt spray test, shall show no more than 1/8 inch creepage, blistering, or loss of adhesion of the paint as measured from the scribe mark. At all other points there shall be no more than a trace of film failure such as blistering, or corrosion, and not more than ten scattered blisters, none larger than 3/16 inch in diameter on the painted surface of the warhead.

5. PACKAGING

5.1 Packaging and marking. Warheads shall be packed and marked in accordance with Dwg. 8837835.

5.2 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Intended use. The warhead assemblies covered by this specification are intended for use in the 2.75 Inch Rocket System.

6.2 ordering data. See MIL-A-48078.

6.3 Inspection equipment. The contractor shall design inspection equipment as required for the Inspection Equipment Lists (EL) referenced on the applicable Equipment Tabulation (ET) in accordance with the instructions of paragraphs 6.3.1 through 6.3.3.

6.3.1 Inspection equipment lists (EL). Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment the numbers of drawings or Federal Stock numbers of existing designs, or codes as indicated in paragraph 6.3.2 Design action required of the contractor is described in paragraph 6.2 of MIL-A-48078. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.3.

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6.3.2 Inspection equipment list codes. The inspection equipment will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format or commercial equipment.

MU - SIE design, mandatory for use.

6.3.3 Submission of inspection equipment for design approval. See 6.2.3 of MIL-A-48078. Submit equipment designs as required to Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAF-I(D), Dover, NJ 07801.

6.4 First article. When a first article is required for inspection and approval, see 3.10 and 4.3.

6.5 Source control and spacification control drawings. Certificate of conformance from the supplier certifying that the item complies with the drawing requirements shall be provided to the government representative.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1340 -A660)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-50504A	WARHEAD, 2.75 INCH ROCKET, M151 PARTS FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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